

Taiwan Renewable Electricity Procurement Workshop — Design and Facilitator Guide v0.1

> PRE-KICKOFF DRAFT。AMAT stakeholder design、target、site、volume、policy 與 preferences 均為 TBD WITH CLIENT。本文件提供 market education / decision facilitation，不是法律意見。

1. Workshop objective

工作坊結束時，參與者應能：

- 用共同語言描述台灣企業綠電 procurement path 中的 buyer、generator / developer、renewable electricity retailer、grid / wheeling 與 T-REC interfaces。
- 區分 physical electricity、commercial contract / settlement、certificate / claim evidence 三種 flow。
- 理解 onsite、direct supply、wheeling / retailer-enabled PPA、certificate-only 等 options 的 volume、tenor、price、flexibility、execution complexity 與 risk 差異。
- 對 AMAT 應優先分析的 options、decision criteria、inputs、owners 與 next gates 形成初步共識。

Workshop 不以完成所有法規 / 合約教學為目的，也不在沒有 AMAT 資料時預先選定供應商或結論。

2. Participant / audience design

Role hypothesis	Why needed	Must attend	Useful segment	Pre-work
Taiwan sustainability / RE100 owner	target、claim、reporting 與 success definition	Opening、options、decisions	all	target / policy questions
Procurement	market process、supplier engagement、commercial comparability	flows、trade-offs、decisions	slides 4–15	existing process / disclosure rules
Facilities / site energy	meter / account、load、onsite / wheeling execution	flows、constraints、data asks	slides 4–15	site / data constraints
Finance	cost baseline、budget、settlement / credit / volatility	trade-offs、criteria、decisions	slides 8–15	evaluation horizon / cost needs
Legal	NDA、market disclosure、contract / risk allocation、legal advice	boundary、risks、next gates	slides 6–15	legal process / counsel role

Additional participant hypotheses

Role hypothesis	Why needed	Must attend	Useful segment	Pre-work
Regional / global sustainability	policy alignment 、 reporting / claim consistency	why 、 options 、 decisions	slides 1–5, 10–15	global policy constraints
Regional / global procurement	mandate 、 supplier / approval architecture	criteria 、 gates	slides 8–15	approval / vendor rules
Executive sponsor / decision maker	priority 、 trade-off 、 resources 、 gate decisions	opening and final decisions	slides 1–3, 11–15	success / deadline statement

以上全為 HYPOTHESIS 。 Sponsor 應在邀請前決定 required / optional segments ， 避免一律邀請全程 。

3. Recommended format

120-minute decision workshop

Time	Module	Purpose	Output
0–10	Why this matters / objectives	建立共同 decision context	confirmed objectives / parking lot
10–35	Market participants and three flows	共同語言	clarified terms / misunderstandings
35–60	Procurement paths	理解可行結構與 interfaces	options to assess / exclude from MVP
60–70	Break / pulse check	收斂 open questions	prioritised questions
70–95	Trade-offs and constraints	設定 1.5 criteria	draft decision criteria / weights
95–112	AMAT implications and data needs	連到 1.4–1.6	owners / inputs / due dates
112–120	Decisions and close	確認 next gates	decisions / actions / open issues

若只允許 90 分鐘 ， 縮短 option deep dive ， 不刪除決策與 data owner 段落 。

4. Market structure teaching frame

Participants

- Corporate buyer ： 定義需求 、 contracting / credit 、 internal approvals 、 claim / reporting 、 implementation owner 。
- Renewable generator / developer ： project / asset 、 generation 、 COD / availability 、 delivery / curtailment / performance evidence 。
- Renewable electricity retailer ： 可聚合 / 購售 、 contract interface 、 settlement / service 與市場接觸角色 ； 實際責任依 proposal / contract 驗證 。

- Taipower / grid interface：在轉供 / 併網型直供情境涉及 network / operational process 與電量資料；current process 以官方規章為準。
- T-REC Center / BSML：憑證申請、查驗、電量累計、讓與、使用 / 宣告的制度介面。
- Advisors / legal counsel / verifiers：方法、商務、法律、查驗角色須分清。

Three flows

1. Electricity / operational flow：generation → grid / direct connection → buyer meter / account；涉及 site、meter、timing 與 actual delivery。
2. Contract / money flow：PPA / retailer / service / network / settlement agreements；涉及 price、volume、tenor、credit、default、change、tax / accounting。
3. Certificate / claim flow：generation verification → certificate issuance → transfer / ownership → use / claim record；不得只因簽了 electricity contract 就假設 claim evidence 完整。

Facilitator 應要求參與者對每個 option 分別指出三種 flow 與 owner。

5. Procurement option frame

Option archetype	Strategic use	Strengths to test	Constraints / risks to test	Client questions
Onsite self-generation / self-use	site-level visible action、部分 load coverage	site proximity、potential operational / reputational benefit	roof / land、capex / opex、generation profile、site tenure、certificate treatment	哪些 sites / loads / ownership 可行？
Direct supply	generator-to-user direct line in qualifying structure	potentially clearer physical linkage	location / interconnection / capex / operational feasibility	是否有 site-specific condition 值得研究？
Offsite wheeling PPA	medium / long-term core volume	scale、tenor、technology / project linkage	project maturity、COD、profile、wheeling、curtailment / delivery、credit、contract complexity	需要多少 firmness / flexibility？
Retailer-enabled procurement	access / aggregation / interface simplification	portfolio / operational support potential	pass-through terms、supplier dependence、transparency、credit / settlement	AMAT 要 single interface 還是 asset-level control？
Certificate-only / unbundled T-REC	gap-filling / timing flexibility subject to policy	speed、smaller volume、limited physical integration	eligibility / additionality / vintage / availability / claim treatment / price	global policy 如何看待？用於何種 gap？
Portfolio combination	balance timing、risk、volume、technology	diversification / staged execution	governance、double count、data / reconciliation complexity	哪些 constraints must be satisfied together？

表格是 analysis framework，不是 AMAT recommendation。

6. Trade-off axes

使用 1–5 scale 前先定義量尺；不在 workshop 中無依據打分。至少討論：

- Volume / coverage / profile fit
- Timing / COD / contract start
- Tenor / renewal / exit / flexibility
- Price structure / escalation / settlement / budget volatility
- Technology / asset / project maturity / concentration
- Delivery / curtailment / generation / counterparty risk
- Wheeling / meter / certificate / implementation complexity
- Claim / RE100 relevance under client policy
- Legal / credit / guarantee / change / termination complexity
- Internal resource and approval burden

7. Taiwan market constraints discussion

避免以一句「市場缺電 / 缺綠電」取代 evidence。將限制拆成：

- Asset / supply availability and project maturity
- Network / wheeling / meter / operational process
- Certificate issuance / transfer / claim timing
- Contracting / credit / guarantee and counterparty capacity
- Volume / profile / technology / site fit
- Internal approval / data / timeline constraints

對每一項區分 official rule、supplier statement、public evidence、Optimum analysis、client decision。

8. Discussion questions by module

Why / success

- 此 project 最重要是 target compliance、execution certainty、cost visibility、portfolio resilience 或其他？
- 哪個 decision / deadline 不能錯過？

Flows / roles

- 哪個 function 擁有 electricity data、contract、certificate / claim、budget 與 site implementation？
- 哪個 interface 最容易被誤以為「別人會處理」？

Options / trade-offs

- 哪些 option 必須納入 1.5 base case / sensitivity？
- 哪些 constraints 是 hard constraint，哪些只是 preference？
- 更重視 scale / long-term economics，還是 speed / flexibility？

Decisions / inputs

- 1.4 residual gap 口徑誰批准？
- 1.5 criteria / weights 與 risk tolerance 誰批准？
- 進 2.x 前，identity / volume / site / timing 可揭露到什麼程度？

9. Facilitator run-of-show

Before

- Sponsor alignment：objective、decisions、participants、sensitive topics、what not to decide。
- Send 2-page pre-read / 5-question survey；不要先發完整法規資料包。
- Assign facilitator、market SME、note / decision recorder、timekeeper。
- 準備 decision log、parking lot、action log 與 truth-status legend。

During

- 開場先確認「今日要決定 / 不決定」。
- 每個 fact 標 source / date；遇到未確認 current detail 標 RESEARCH REQUIRED，不即席猜。
- 每個 option 先畫三種 flow，再談 price / recommendation。
- 出現 disagreement 時記錄 decision owner、needed evidence、latest-needed date。
- 10 分鐘一次 pulse check；timebox deep dive，移入 parking lot。

After — within 48 hours

- 發送 workshop decision / action / assumption / question record。
- 將 data asks 寫回 1.2 tracker。
- 將 gap boundary 寫入 1.4 model assumptions。
- 將 option / criteria / constraints 寫入 1.5 scenario design。
- 將 stakeholders / approvals / deadlines 寫入 1.6 roadmap。
- 需研究的 current market / legal detail 進 source / research log。

10. Expected decisions / learnings template

Item	Decision / learning	Status	Owner	Due	Downstream
Success priority	TBD WITH CLIENT	Open	Sponsor	Workshop close	1.5 weights / 1.6
Target / gap boundary	TBD WITH CLIENT	Open	Sustainability owner	G1	1.4
Options for first-pass scenario	TBD WITH CLIENT	Open	Steering group	G2	1.5
Hard constraints	TBD WITH CLIENT	Open	Cross-functional	G2	1.5 / 2.x
Market disclosure boundary	TBD WITH CLIENT	Open	Procurement / Legal	Before 2.2	2.x
Data / evidence gaps	From 1.2 / workshop	Open	Named owners	G1 / G2	1.4–1.6

11. Source and research protocol

Official reference set — accessed 2026-09-27

- [T-REC documents and current application guidance](<https://www.trec.org.tw/documents>) — official operational guides, including self-use and direct / wheeling facility certificate guidance.
- [Implementation Regulations Governing Renewable Energy Certificates](<https://www.trec.org.tw/index.php/en/regulations>) — official descriptions of certificate application, issuance / accrual, transfer and claim use.
- [Taipower wheeling and grid-connected direct supply operating rules](<https://service.taipower.com.tw/powerwheeling/static/file/1130318%E8%BD%89%E7%9B%B4%E4%BE%9B%E7%87%9F%E9%81%8B%E8%A6%8F%E7%AB%A0.pdf>) — official operating reference; check latest revision before client delivery.
- [Electricity Act copy published by Taipower](<https://service.taipower.com.tw/powerwheeling/static/file/1130718%E9%9B%BB%E6%A5%AD%E6%B3%95.pdf>) — legal definitions / authorities; formal legal interpretation remains client counsel's role.

Use rule

- Every current-process claim in client material gets source / publication or access date.
- Check latest version immediately before workshop; webpage or PDF may change.
- Do not present supplier statements as official rules or legal advice.
- Public market data 、 price 、 availability 、 project claims not yet researched remain RESEARCH REQUIRED 。